THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING

DEPARTMENT OF CIVIL ENGINEERING

CIVL1113: ENGINEERING MECHANICS AND MATERIALS

May 16, 2026 TIME: 2:30pm – 5:30pm

Candidates are permitted to bring to the examination ONE sheet(s) of A4-sized paper with printed/written notes on both sides.

Answer ALL **5 (FIVE)** questions from Section A and B.
Marks for sub-questions (if any) are given.

Use a separate answer book for each section. Use the Blue Answer Book for Section A and the Green Answer Book for Section B.

Use of Calculators:

Only approved calculators announced by the Examinations Secretary can be used in this examination. It is your responsibility to ensure that your calculator is functioning properly during the examination. You must record the name and model of the calculator used on the front page of the examination script.

SECTION A

Q1 (a) As shown in Figure 1(a), the beam ABCD as shown in the figure has overhangs at each end and carries a uniform load of intensity q.

(i) For what ratio b/L will the bending moment at the mid-point of the beam be zero. Draw the bending moment diagram with pertinent values for this case. **(5 marks)**

(ii) For what ratio b/L will the bending moment at the mid-point of the beam is positive and equal to the magnitude of negative moment at B section. Draw the bending moment diagram with pertinent values for this case. **(5 marks)**

(b) A plane frame is constructed by a beam and a column with a rigid connection at B. The external load P is inclined and angle α=30 degrees. Figure 1(b) shows the problem.

(i) Draw the shear force diagram (SFD) of the frame and axial force diagram with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. Note that SFD and axial force diagram of all components need to be plotted. **(5 marks)**

(ii) Draw the bending moment diagram (BDM) of the frame with pertinent values. Noe that bending moment should be drawn on the **tensile** side. Note that BMD of all components need to be plotted. **(5 marks)**

Q2 (a) Determine the maximum tensile stress and maximum compressive stress due to the load P acting on the simple beam AB as shown in Figure 2(a). The yield strength is obtained from tensile test as 16MPa. The safety factor is 1.0. Data are as follows: P=6kN, L=4 m, d=0.1m, flange width b=60mm, web thickness t=4mm, total height h=200mm, and height of web is h_1=190mm.

(i) Check the maximum normal stress and maximum shear stress of the beam. **(5 marks)**

(ii) Check the safety of the beam using Tresca criterion, which is also known as maximum shear stress criterion **(5 marks)**

(b) A point is subject to plane stress state as shown in Figure 2(b). x axis is horizontal and pointing to the right. y axis is vertical upward. The von Mises criterion is used for the material. The yield strength is obtained from tensile test as 175 MPa. Safety factor is 1.0.

(i) Determine the stress acting on an element oriented at an angle θ=45 degrees from the x axis. Rotating counterclockwise from x axis is positive. **(5 marks)**

(ii) Check the safety of the point using von Mises criterion. **(5 marks)**

Q3 (a) A rod of length L=150 mm is to be twisted by torques *T* as shown in Figure 3(a) until the angle of rotation between ends of the rod is 2 degrees. The elastic modulus is 200 GPa and the Poisson ratio is 0.3. If the allowable shear strain in the rod material is 0.0006 rad, what is the maximum permissible diameter of the rod. **(5 marks)**

(b) The column ABC shown in Figure 3(b) is subject to a downward load P. Both the top C point and the bottom A point are spherical hinges. B point only restricts in-plane deformation but permit out-of-plane deformation. Material is steel with an elastic modulus of 200GPa. b=25 mm and h=45 mm. The Poisson ratio is 0.3. The cross-section of the members is as shown in Figure 3(b). The distance between the A and C is L=30 m. Assume the material will not fail due to strength issue. Consider buckling in-plane and out-of-plane. For buckling design, safety factor is 1.0 for this problem. Determine the allowable value of the load *P*. **(5 marks)**

FIGURES

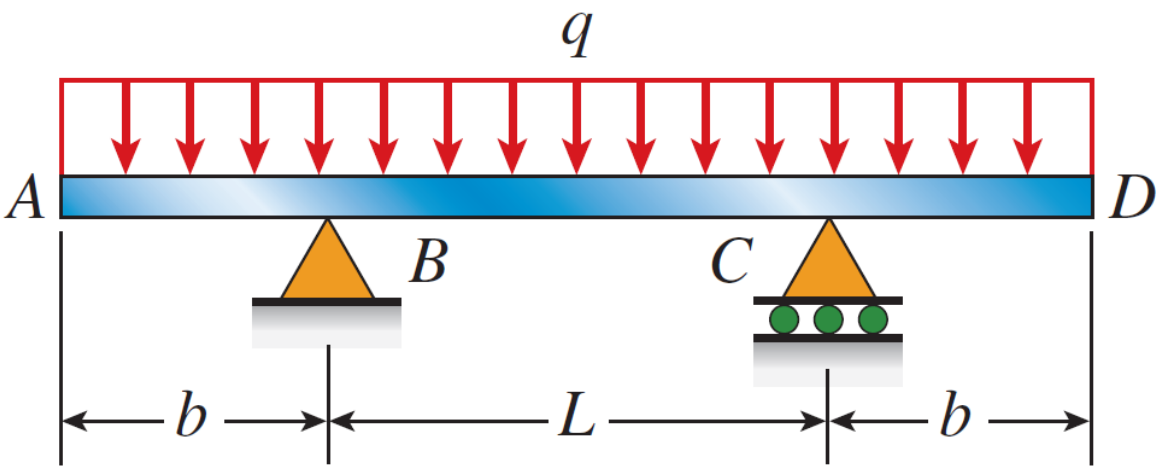


Figure 1(a). Beam

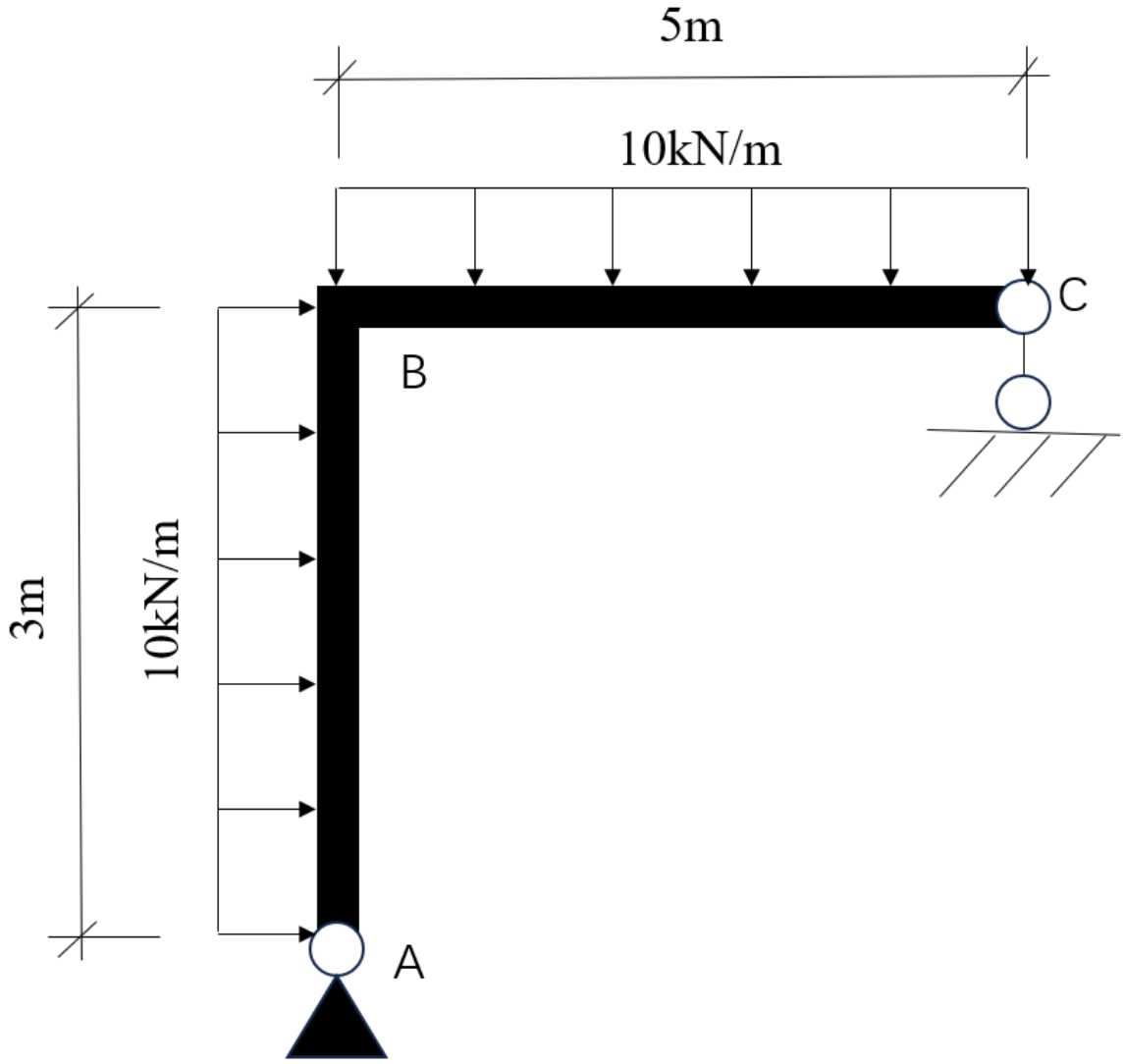


Figure 1(b). Plane frame

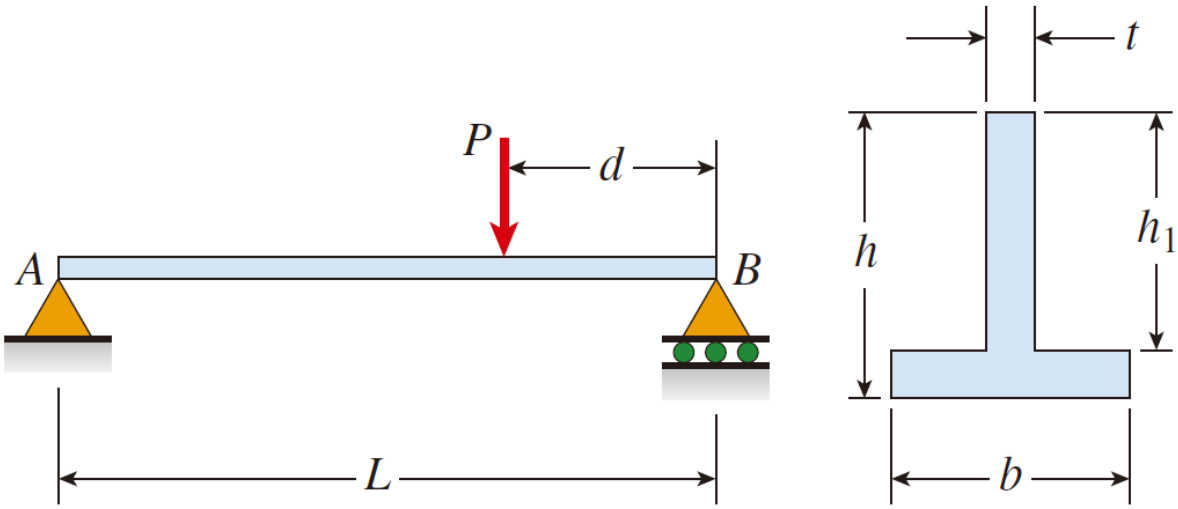


Figure 2(a). Simple beam section

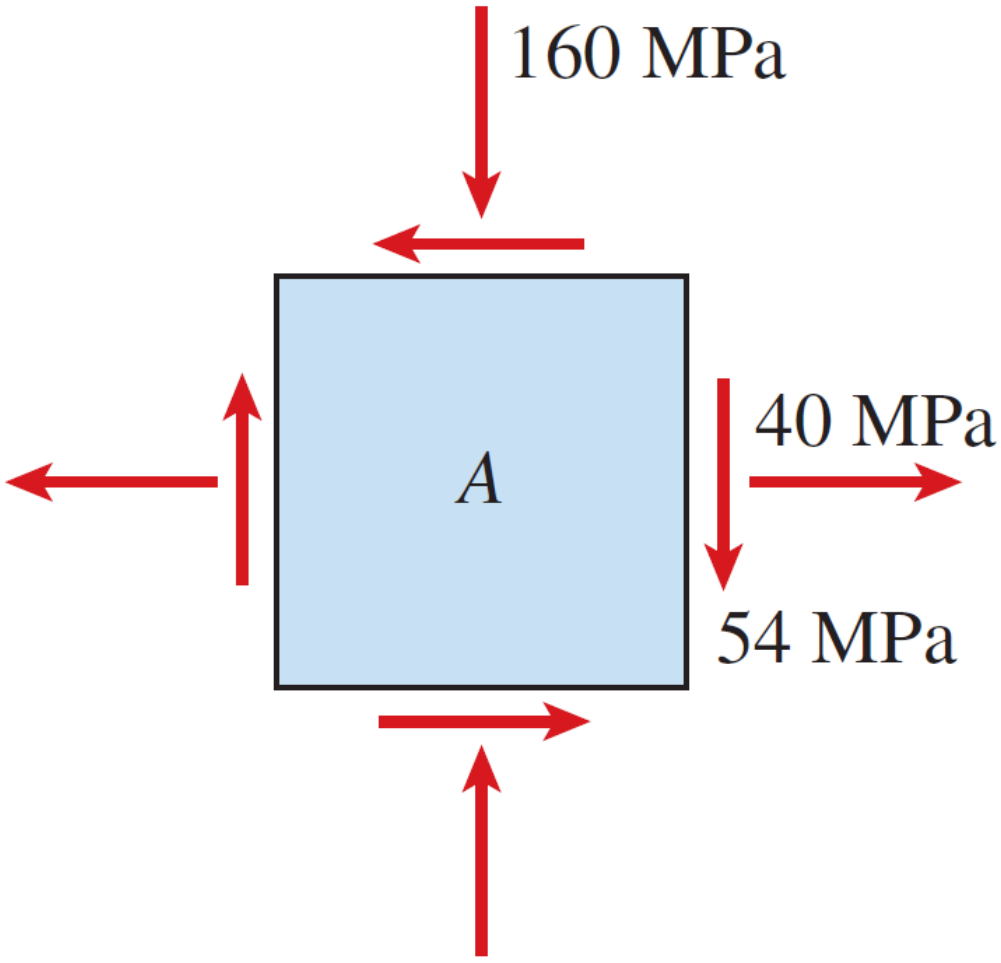


Figure 2(b). Plane stress at a point

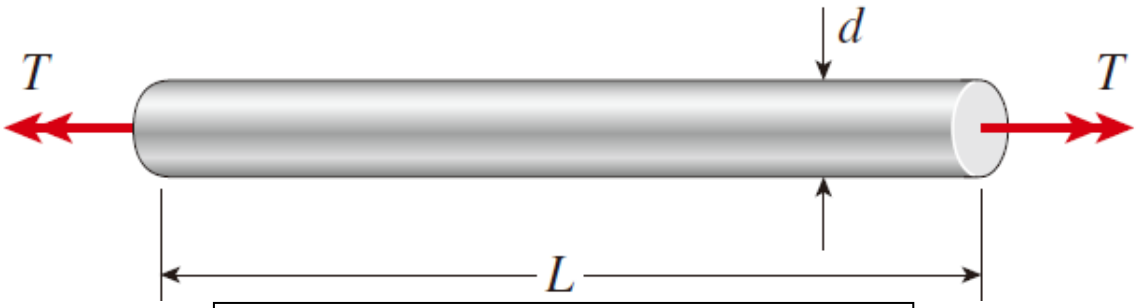


Figure 3(a). Rod under pure torsion

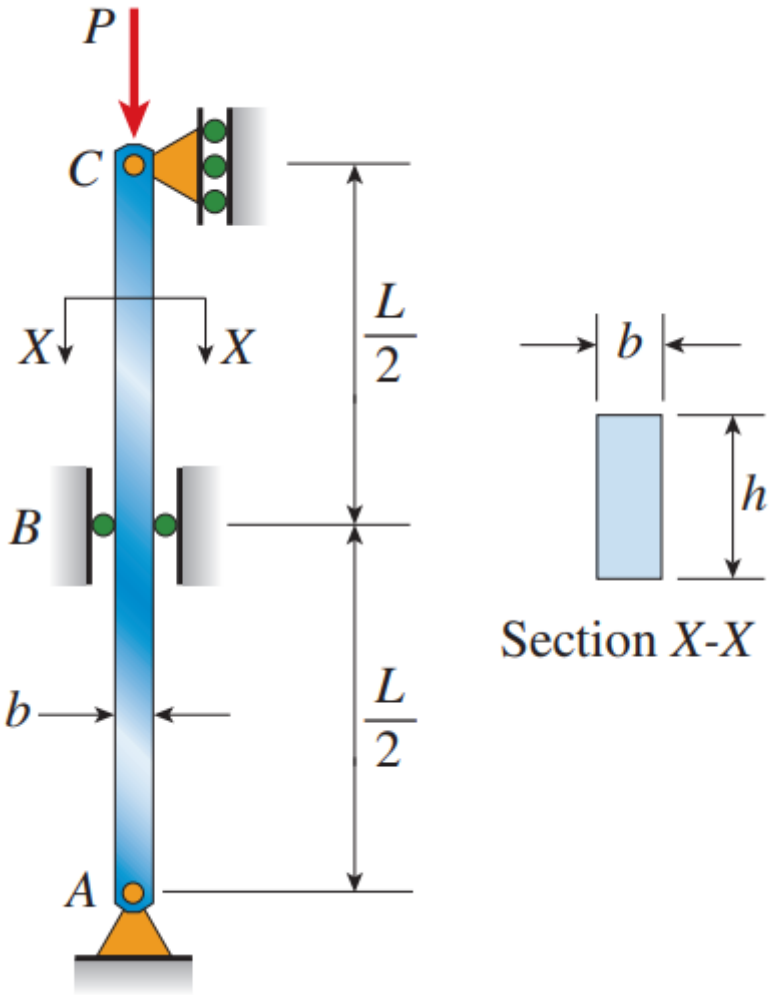


Figure 3(b). Column

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